Correction notice

Genomic Identity of the Sima de los Huesos Hominins: Convergent Evidence from Mitochondrial and Nuclear DNA — version 2

Emilio García-Morán · 21 October 2026 · Supersedes version 1 (doi:10.5281/zenodo.19389351, April 2026) and the October 2026 manuscript *Phylogenetic placement of Sima de los Huesos hominins revisited*.

Summary

Part	Status
Mitochondrial DNA	**Corrected.** With explicit *Pan troglodytes* rooting, Sima is sister to Denisova, as reported by Meyer et al. (2014). The “basal Sima” placement is withdrawn.
Description of Meyer et al. (2014)	**Corrected.** They used a Bayesian analysis (MrBayes, GTR+I+Γ; ~120 mtDNAs including 46 chimpanzee and bonobo genomes; D-loop removed), not an unrooted Neighbor-Joining tree.
Nuclear DNA (D-statistics, distances)	**Under verification.** Withdrawn as results until the original ANGSD outputs are recovered and checked.
Phase 2 mtDNA analyses	**Withdrawn.** Input data included non-hominin sequences.

1. Mitochondrial DNA

What was claimed. Version 1 described Sima de los Huesos as the basal sister lineage to (Denisova + Neandertal), with 93% IQ-TREE bootstrap support, and interpreted the Sima–Denisova affinity reported by Meyer et al. (2014) as an artefact of an unrooted analysis.

What went wrong. The IQ-TREE run was performed without specifying the outgroup; its own output states that the tree is unrooted and that Sima is only *drawn* at the root. The 93% value supports the split {Sima, Denisova} | {Neandertals, *Pan*}, which, rooted on *Pan*, is the Sima + Denisova clade. Reading Sima as basal was a misreading of an unrooted tree, and the published tree figures were typed by hand rather than drawn from the output. The original computations were correct: the March 2026 RAxML-NG bootstrap replicates already supported Sima + Denisova in 98.4%.

Corrected result. Six complete mitochondrial genomes (Sima Femur XIII KF683087, Denisova 3, Vindija 33.16, Feldhofer 1, Mezmaiskaya 1, *Pan troglodytes*), re-analysed from GenBank with automatic validation of every record, MAFFT 7.526, IQ-TREE 3.0.1 and RAxML-NG 1.2.2, all rooted on *Pan*. Bootstrap support (% of 1,000 replicates):

Alignment	Analysis	Sima + Denisova	Sima + Neandertals	Denisova + Neandertals
full	IQTREE TPM2u+F+G4 UFBoot	**98.0**	2.0	0.0
full	IQTREE HKY+G4 standard bootstrap	**92.9**	7.1	0.0
full	RAxML GTR+G standard bootstrap	**97.9**	2.1	0.0
no D-loop	IQTREE HKY+F+G4 UFBoot	**97.4**	1.2	1.4
no D-loop	IQTREE HKY+G4 standard bootstrap	**98.8**	0.4	0.8
no D-loop	RAxML GTR+G standard bootstrap	**99.6**	0.3	0.1

Joint topology tests (10,000 RELL replicates) reject “Sima basal” in all four model/alignment combinations; Sima + Denisova has the highest likelihood in all four:

Alignment	Model	Hypothesis	ΔlogL	p-AU
full	HKY+G4	H0 Sima basal (withdrawn claim)	10.52	**0.011**
full	HKY+G4	H1 Sima + Denisova	0.00	0.946
full	HKY+G4	H2 Sima + Neandertals	8.74	0.070
full	TPM2u+F+G4	H0 Sima basal (withdrawn claim)	12.35	**0.009**
full	TPM2u+F+G4	H1 Sima + Denisova	0.00	0.962
full	TPM2u+F+G4	H2 Sima + Neandertals	10.68	**0.049**
no D-loop	HKY+G4	H0 Sima basal (withdrawn claim)	6.39	**0.038**
no D-loop	HKY+G4	H1 Sima + Denisova	0.00	0.963
no D-loop	HKY+G4	H2 Sima + Neandertals	6.39	**0.038**
no D-loop	HKY+F+G4	H0 Sima basal (withdrawn claim)	6.39	**0.038**
no D-loop	HKY+F+G4	H1 Sima + Denisova	0.00	0.963
no D-loop	HKY+F+G4	H2 Sima + Neandertals	6.39	**0.038**

Figure 1. Rooted mitochondrial phylogeny: Sima de los Huesos is sister to Denisova

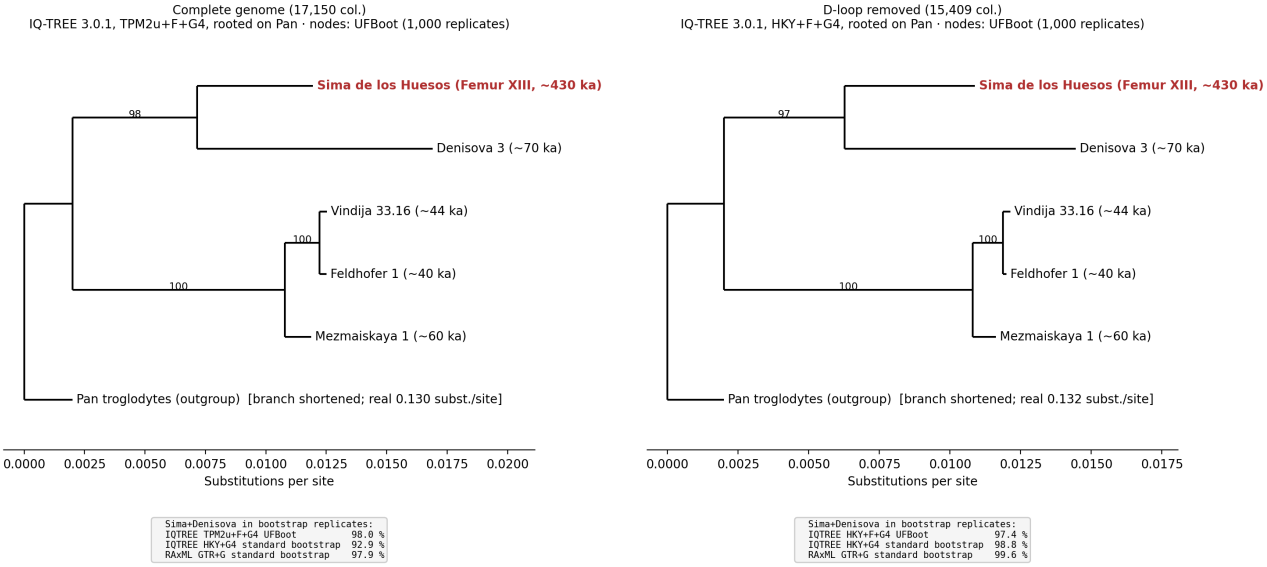


Figure 1. Rooted mitochondrial phylogeny drawn directly from the IQ-TREE output (UFBoot values at nodes; *Pan* branch shortened for display, real length indicated).

This agrees with Meyer et al. (2014) and with Posth et al. (2017), who also rooted on chimpanzee. The mito-nuclear discordance is explained by the replacement model: Sima retains the ancestral, Denisovan-like mtDNA of the early Neandertal lineage, later replaced in Neandertals by an introgressed African mtDNA between ~413 and ~268 ka (Posth et al. 2017); a parallel replacement is documented for the Y chromosome (Petr et al. 2020).

2. Phase 2 analyses (withdrawn)

Accession numbers in the phase 2 scripts were typed by hand and never checked against the records GenBank returned. Ten are not hominin mitochondrial DNA:

Used as	Accession	Actual GenBank record
Oase 1, Tianyuan, Kostenki 14, Vindija 33, Denisova 4/8	KX198073/74/75/78/79	*Agaricus* (fungus), EF1α gene
“Sima SH5”	KY751401	*Henneguya tucunarei* (fish parasite), 18S rRNA
Ust’-Ishim	KP295291	*Phytophthora castaneae* (oomycete), HSP90
Ust’-Ishim	KC758896	*Ostrinia scapulalis* (moth), ABC transporter
“Denisova 2”	NC_024829	*Nephroselmis astigmatica* (green alga), chloroplast genome
“Neandertal”	EU260658	Simian immunodeficiency virus, partial *tat* gene

Phase 2 results, including the “6–0” Sima–Denisova shared-site count, are withdrawn. The six genomes of the corrected analysis are unaffected; their GenBank titles and sequence hashes are now validated before alignment.

3. Nuclear DNA (under verification)

The nuclear section of version 1 (chromosome 1 only; D = 0.7425, Z = 30.6; pairwise distances) cannot yet be reproduced: the ANGSD commands and outputs were not archived with the code, and the figure scripts contain hand-typed values. These numbers are withdrawn as results while the original outputs are recovered. A genome-wide, fully reproducible nuclear analysis is planned as a separate project. The taxonomic proposal of version 1 is withdrawn with it.

4. Data, code and reproducibility

Repository: https://github.com/EmilioGarciaMoran/sima-genomics (CORRECTION.md; analysis in mtDNA_repro/). One command, bash mtDNA_repro/run_all.sh genbank, reproduces the analysis from GenBank to the figure (~16 min on 2 cores). An independent clean-clone run reproduced sequences, alignments and all bootstrap frequencies bit for bit, and the author’s local run downloading from GenBank returned identical sequence hashes. Superseded scripts and manuscripts remain in the repository with notices. This PDF was generated from commit 420e51c by tools/build_correction_pdf.py.

References

Meyer M. *et al.* (2014) A mitochondrial genome sequence of a hominin from Sima de los Huesos. *Nature* 505:403–406. doi:10.1038/nature12788

Meyer M. *et al.* (2016) Nuclear DNA sequences from the Middle Pleistocene Sima de los Huesos hominins. *Nature* 531:504–507. doi:10.1038/nature17405

Posth C. *et al.* (2017) Deeply divergent archaic mitochondrial genome provides lower time boundary for African gene flow into Neanderthals. *Nat. Commun.* 8:16046. doi:10.1038/ncomms16046

Petr M. *et al.* (2020) The evolutionary history of Neanderthal and Denisovan Y chromosomes. *Science* 369:1653–1656. doi:10.1126/science.abb6460